



Uttar Pradesh Power Corporation Limited (UPPCL)
&
Uttar Pradesh State Load Dispatch Centre
(UPSLDC)

SUMMER TRAINING

A report submitted in the partial fulfilment of the requirements for Summer / Industrial Training at
Uttar Pradesh Power Corporation Limited

By

Akash Yadav (2410013125039)
(STUDENT B.TECH ELECTRONICS AND COMMUNICATION ENG.)



University of Lucknow

Akash Yadav , student of university of Lucknow would like to extend my sincere gratitude to the management of Uttar Pradesh State Load Dispatch Centre (UPSLDC), Lucknow, for providing me with the opportunity to undergo industrial training in one of the most important power system operation centres of Uttar Pradesh.

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Akash Yadav

B.Tech (ECE)

Faculty of Engineering & Technology

University of Lucknow

ABSTRACT

Industrial training plays an important role in bridging the gap between classroom learning and industrial applications. This internship was carried out at Uttar Pradesh State Load Dispatch Centre (UPSLDC), Lucknow, which is responsible for the real-time monitoring, control, scheduling, and secure operation of the power transmission network of Uttar Pradesh.

The primary objective of this training was to understand the practical implementation of Supervisory Control and Data Acquisition (SCADA) and Energy Management System (EMS) used in modern power system operation. During the internship, practical exposure was gained in SCADA architecture, communication systems, optical fibre communication, Remote Terminal Units (RTUs), Intelligent Electronic Devices (IEDs), GPS time synchronization, network security, battery backup systems, and generation scheduling.

The internship also provided an opportunity to study communication equipment such as routers, switches, firewalls, fibre optic systems, SDH equipment, Fusion Splicer, Optical Time Domain Reflectometer (OTDR), and communication monitoring software. The functioning of the SCADA Control Room and real-time monitoring of substations, transmission lines, and generating stations were observed.

This report presents a comprehensive overview of the knowledge acquired during the internship along with practical observations, technical concepts, and the role of modern automation technologies in ensuring reliable, secure, and efficient operation of the electrical power grid.

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Introduction to the Indian Power System

Electricity is one of the most essential forms of energy and plays a vital role in the economic and industrial development of a country. The Indian Power System is one of the largest synchronized power systems in the world, consisting of generating stations, transmission networks, substations, distribution systems, and millions of consumers. It operates continuously to ensure a reliable and uninterrupted supply of electrical energy across the nation.

The power system mainly consists of four stages:

- Power Generation
- Power Transmission
- Power Distribution
- Power Consumption

Electricity generated at generating stations is transmitted over Extra High Voltage (EHV) transmission lines to minimize transmission losses. It is then stepped down through substations before being supplied to domestic, commercial, and industrial consumers.

Maintaining a continuous balance between power generation and electricity demand is one of the most challenging tasks in power system operation. Any imbalance may result in frequency deviation, voltage instability, equipment damage, or even large-scale blackouts. Therefore, a centralized monitoring and control system is essential for secure and reliable operation of the grid.

Structure of the Indian Power System

The Indian power system is organized into different operational levels.

National Load Dispatch Centre (NLDC)



Regional Load Dispatch Centre (RLDC)



State Load Dispatch Centre (SLDC)



Transmission Substations



Distribution Companies (DISCOMs)



Consumers

Each level performs a specific function to ensure reliable power transmission and efficient grid operation

National Grid

India operates a unified national grid in which all regional grids are interconnected and synchronized at a nominal frequency of 50 Hz. The National Grid enables power exchange between different regions, improves reliability, enhances system security, and facilitates economic dispatch of electricity.

Voltage Levels in India

The major transmission voltage levels used in India are:

Voltage Level
Application
765 kV
Extra High Voltage Transmission
400 kV
Inter-State Transmission
220 kV
State Transmission Network
132 kV
Sub-Transmission
33 kV
Primary Distribution
11 kV
Distribution
415/230 V
Consumer Supply

Importance of Load Dispatch Centres

Load Dispatch Centres are responsible for monitoring, controlling, and coordinating the operation of the electrical power system.

Their major responsibilities include:

- Grid monitoring
- Power scheduling
- Frequency control
- Voltage regulation

- Outage coordination
- Renewable energy integration
- Grid security
- Emergency restoration

In Uttar Pradesh, these responsibilities are performed by the Uttar Pradesh State Load Dispatch Centre (UPSLDC).

Uttar Pradesh State Load Dispatch Centre (UPSLDC)

Overview

The Uttar Pradesh State Load Dispatch Centre (UPSLDC) is the apex organization responsible for real-time operation of the state's transmission network. It continuously monitors power generation, transmission lines, substations, transformers, renewable energy sources, and power demand across Uttar Pradesh.

UPSLDC functions under the Electricity Act, 2003 and acts as the central coordinating agency for maintaining grid stability and ensuring reliable power supply throughout the state.

The control centre operates 24 hours a day and 365 days a year, making it one of the most critical facilities in the state's electrical infrastructure.

Vision of UPSLDC

The primary vision of UPSLDC is to ensure:

- Reliable grid operation
- Safe power transmission
- Efficient utilization of available generation
- Secure communication infrastructure
- Stable frequency and voltage
- Integration of renewable energy resources
- Continuous monitoring of the transmission network

Functions of UPSLDC

The major functions performed by UPSLDC include:

Real-Time Monitoring

Continuous monitoring of:

- Voltage
- Current
- Frequency
- Active Power (MW)
- Reactive Power (MVAR)
- Transformer Loading
- Circuit Breaker Status
- Transmission Line Loading
- Bus Voltage
- Using the SCADA system.
- Grid Operation

The control room engineers supervise:

- Transmission network
- Power generation
- Load demand
- Line outages
- System disturbances
- Grid restoration
- Generation Scheduling
- Generation scheduling includes:
 - Thermal Power Stations
 - Hydro Power Stations
 - Solar Power Plants
 - Independent Power Producers (IPPs)
 - Central Sector Generation
- The objective is to maintain continuous balance between generation and demand.
- Renewable Energy Monitoring
- UPSLDC continuously monitors solar generation integrated into the state grid.
- Renewable energy forecasting and scheduling have become increasingly important due to the rapid growth of renewable installations in Uttar Pradesh.
- Communication Management
- Reliable communication is maintained through:
 - Optical Fibre Communication
 - OPGW
 - SDH Network
 - MPLS
 - Ethernet Network
 - Routers
 - Switches
 - Firewalls
 - GPS Time Synchronization

These communication systems ensure uninterrupted data transfer between substations and the control centre.

Major Departments Observed During Internship

During the industrial training, exposure was provided to several important technical sections of UPSLDC, including:

- SCADA Division
- Energy Management System (EMS)
- Communication Room

- Optical Fibre Communication System
- Network Infrastructure
- Battery Charger & DC Supply System
- GPS Time Synchronization
- Communication Monitoring Software
- Generation Scheduling
- Control Room Operations

SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

Introduction

The Supervisory Control and Data Acquisition (SCADA) system is one of the most important automation technologies used in modern electrical power systems. It enables engineers to monitor, supervise, analyse, and control geographically distributed substations and transmission networks from a centralized control centre.

In a large state like Uttar Pradesh, where hundreds of generating stations, substations, and transmission lines are interconnected, manual monitoring is neither practical nor reliable. SCADA overcomes this challenge by continuously collecting operational data from remote substations and presenting it in real time to system operators.

At Uttar Pradesh State Load Dispatch Centre (UPSLDC), Lucknow, SCADA serves as the backbone of grid operation. It allows operators to monitor the status of the transmission network, identify abnormalities, issue control commands, and maintain a secure and stable power system.

Objectives of SCADA

The major objectives of the SCADA system are:

- Continuous real-time monitoring of the transmission network.
- Collection of operational data from substations.
- Remote supervision and control of field equipment.
- Fast detection of faults and abnormal operating conditions.
- Alarm generation and event recording.
- Reduction of outage restoration time.
- Improvement of grid reliability and operational efficiency.
- Support for Energy Management System (EMS) applications.

Need for SCADA

The electrical transmission network of Uttar Pradesh consists of numerous 400 kV, 220 kV, and 132 kV substations spread across a large geographical area. Monitoring each substation manually would require significant manpower and would not provide real-time visibility.

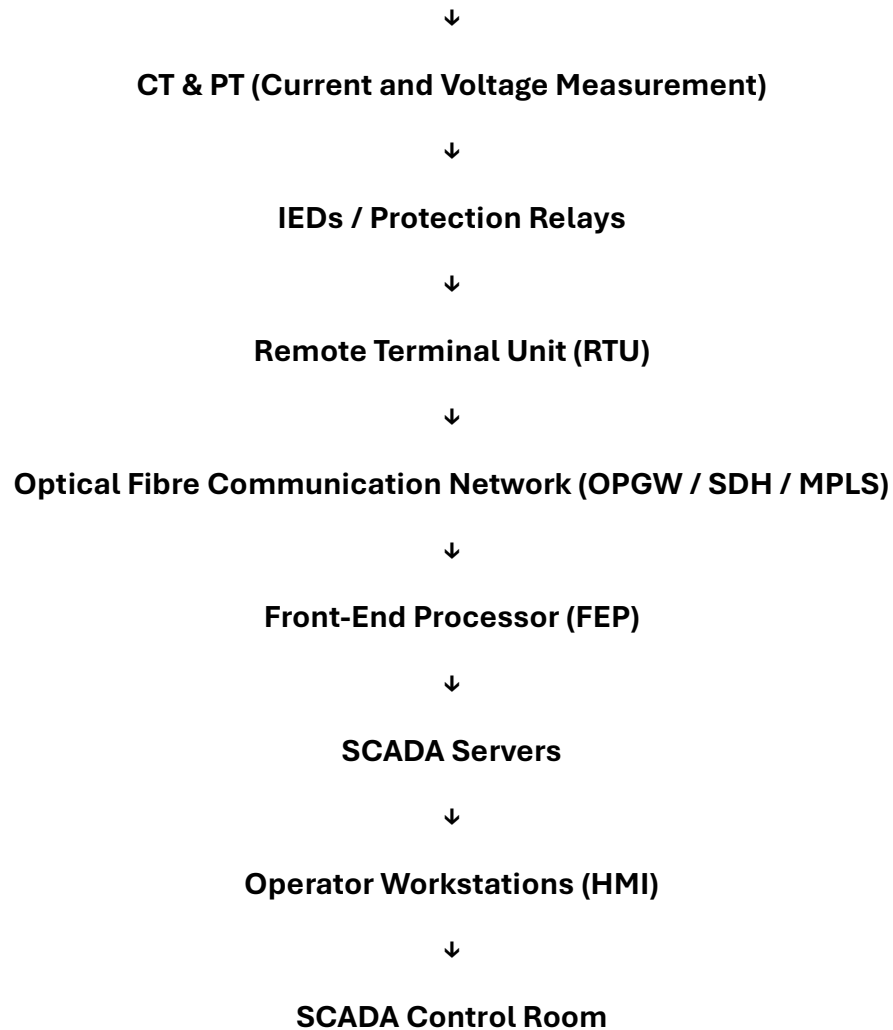
SCADA eliminates these limitations by providing a centralized monitoring platform where operators can instantly observe the status of the complete transmission network. This enables faster decision-making, reduces response time during faults, and improves overall system reliability.

Working Principle of SCADA

The SCADA system works by collecting data from field devices installed at substations and transmitting it to the control centre through a reliable communication network.

The overall process can be summarized as follows:

Power System Equipment (Transmission Lines, Transformers, Circuit Breakers)



This architecture ensures that information from substations reaches the control centre within a few seconds.

Major Components of SCADA

Field Equipment

Field equipment forms the primary source of operational data.

The major field devices include:

- Power Transformers
- Circuit Breakers
- Isolators
- Current Transformers (CTs)
- Potential Transformers (PTs)
- Busbars
- Transmission Lines

These devices continuously provide electrical measurements required for grid monitoring.

Current Transformer (CT)

A Current Transformer is used to reduce high transmission line current into a safe value suitable for measurement and protection.

Functions:

- Current measurement
- Protection relay input
- Metering
- SCADA data acquisition

Potential Transformer (PT)

A Potential Transformer reduces high system voltage into a standard low voltage for measurement and monitoring.

Applications:

- Voltage measurement
- Synchronization
- Metering
- Protection

Intelligent Electronic Device (IED)

IEDs are intelligent microprocessor-based devices installed in substations.

They perform:

- Protection
- Measurement
- Fault recording
- Event recording
- Communication with RTU
- Modern numerical relays themselves function as IEDs.

Remote Terminal Unit (RTU)

The RTU acts as the communication interface between the substation and the control centre.

Functions of RTU include:

- Data acquisition
- Analog input processing
- Digital input monitoring
- Digital output control
- Communication with SCADA servers

Typical data collected by RTUs include:

- Voltage

- Current
- Frequency
- MW
- MVAR
- Breaker status
- Isolator status
- Transformer loading
- Alarm signals

Communication Network Used in UPSLDC

Reliable communication is essential for uninterrupted SCADA operation.

During the internship, the following communication technologies were studied:

- Optical Fibre Communication
- Optical Ground Wire (OPGW)
- SDH Communication Network
- MPLS Network
- Cisco Routers
- D-Link Managed Switches
- Fortinet Firewalls
- GPS Time Synchronization

These technologies ensure secure and reliable transmission of operational data between substations and UPSLDC.

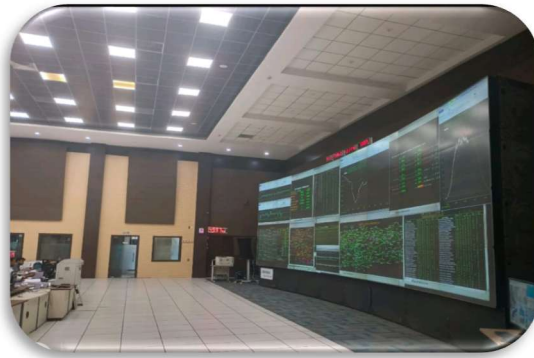
Human Machine Interface (HMI)

The Human Machine Interface is the graphical interface through which operators monitor and control the transmission network.

The HMI displays:

- Single Line Diagram (SLD)
- Bus Voltages
- Frequency
- Power Flow
- Breaker Status
- Transformer Loading
- Line Loading
- Generation Data
- Alarm Messages
- Event Logs

During the internship, the SCADA Control Room displayed these parameters on large video walls and operator consoles, enabling continuous monitoring of the state transmission system.



SCADA Control Room at UPSLDC, Lucknow



Generation Monitoring Dashboard

ENERGY MANAGEMENT SYSTEM (EMS)

Introduction

The Energy Management System (EMS) is an advanced software platform integrated with the SCADA system to support the secure, reliable, and economical operation of the electrical power grid. While SCADA primarily focuses on collecting and displaying real-time data, EMS processes this information to assist operators in analysis, forecasting, scheduling, and decision-making.

At Uttar Pradesh State Load Dispatch Centre (UPSLDC), EMS plays a crucial role in maintaining the balance between electricity generation and demand. It enables system operators to monitor the present condition of the grid, analyse future operating conditions, and take corrective actions to ensure uninterrupted power supply.

The increasing integration of renewable energy sources, such as solar power, has made EMS even more important because renewable generation is variable and depends on weather conditions. EMS helps operators manage these variations while maintaining system reliability.

Objectives of EMS

The primary objectives of the Energy Management System are:

- To maintain continuous balance between power generation and load demand.
- To ensure secure and economical operation of the transmission network.
- To improve overall grid reliability.
- To assist operators in contingency analysis and decision-making.
- To optimize power generation and transmission resources.
- To enhance system security under normal and emergency operating conditions.

Role of EMS at UPSLDC

During the internship, it was observed that EMS supports operators in several important activities, including:

- Monitoring total power generation across Uttar Pradesh.
- Supervising electricity demand in real time.
- Monitoring transmission line loading.
- Managing grid frequency.
- Supporting renewable energy integration.
- Maintaining voltage profile.
- Assisting in power scheduling.
- Providing analytical tools for secure grid operation.

The EMS software receives real-time data from the SCADA system and converts it into meaningful information for system operators.

Working of EMS

The Energy Management System works together with SCADA in the following sequence:

Substations & Generating Stations



RTUs / IEDs



SCADA System



EMS Database



Analytical Applications



Operator Workstations



Decision Making

Unlike SCADA, which mainly monitors field equipment, EMS analyses system conditions and recommends appropriate operational actions.

Major Functions of EMS :

1. Load Monitoring

EMS continuously monitors the electrical load connected to the grid.

It displays:

- State Demand (MW)
- Regional Demand
- Peak Demand
- Minimum Demand
- Real-Time Load Variation

This information helps operators maintain generation-demand balance.

2. Generation Monitoring

The EMS monitors electricity generation from various sources such as:

- Thermal Power Plants
- Hydro Power Plants
- Solar Power Plants
- Independent Power Producers (IPP)
- Central Generating Stations

During the internship, the generation dashboard displayed real-time generation values and scheduled generation for each source.

3. Frequency Monitoring

- Grid frequency is one of the most important parameters monitored by EMS.
- India operates at a nominal frequency of 50 Hz.
- Frequency decreases when demand becomes higher than generation and increases when generation exceeds demand.
- EMS continuously displays system frequency and helps operators take corrective actions whenever necessary.

4. Voltage Monitoring

- The EMS continuously supervises:
- Bus Voltages
- Transformer Voltages
- Transmission Line Voltages
- Maintaining voltage within permissible limits is essential for stable system operation.

5. Alarm Management

Whenever abnormal operating conditions occur, EMS generates alarms for operators.

Typical alarms include:

- Frequency Violation
- Voltage Violation
- Line Overloading
- Transformer Overloading
- Communication Failure
- Generation Loss
- Equipment Trip
- These alarms help operators respond quickly to maintain system stability.
- Historical Data Storage

EMS maintains a historical database containing:

- Load Curves
- Frequency Trends
- Generation Records

- Voltage Profile
- Alarm History
- Event Logs

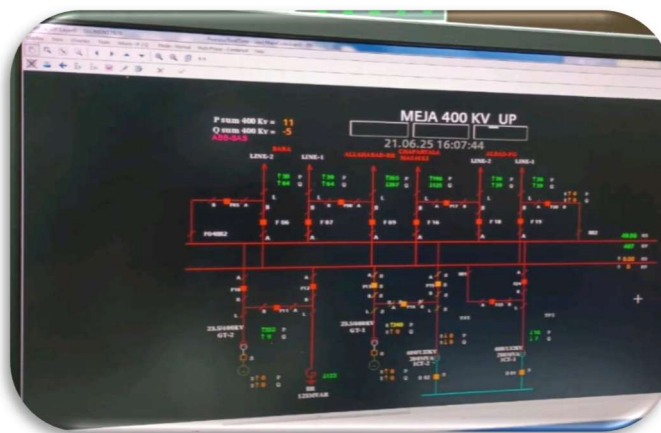
This data is useful for analysis, planning, and operational improvements.

Practical Observation During Internship

During the internship at UPSLDC, the EMS display screens provided valuable information regarding the real-time condition of the power system. Large display panels showed total state generation, demand, frequency, and contribution from thermal, hydro, solar, and independent power producers.

The system also displayed scheduling information and network status, enabling operators to monitor the performance of the entire transmission system from a single control room.

The integration of EMS with SCADA demonstrated how real-time data collected from substations can be transformed into meaningful operational information that supports reliable and efficient grid management.



EMS Generation Monitoring Dashboard

The Energy Management System serves as the analytical and decision-support layer above SCADA. It transforms raw field data into actionable information, enabling UPSLDC engineers to maintain grid stability, optimise generation, monitor renewable energy, and ensure secure operation of the Uttar Pradesh power system.

SCADA Control Centre

Introduction

The SCADA Control Centre is the operational heart of the Uttar Pradesh State Load Dispatch Centre (UPSLDC). It is the centralized facility from which engineers continuously monitor, supervise, and coordinate the operation of the state's transmission network. The control centre operates 24 hours a day, 7 days a week, ensuring the reliable, secure, and economical operation of the electrical grid.

During the industrial internship at UPSLDC, the SCADA Control Centre was one of the most significant sections visited. The centre was equipped with high-performance servers, operator workstations, large display walls, communication systems, and advanced software for monitoring substations and transmission lines across Uttar Pradesh.

Purpose of the SCADA Control Centre

The main objective of the SCADA Control Centre is to provide a centralized platform for real-time monitoring and operational control of the state transmission system.

Its major responsibilities include:

- Monitoring the complete transmission network.
- Supervising generating stations.
- Maintaining grid frequency within permissible limits.
- Monitoring transmission line loading.
- Supervising transformer operation.
- Coordinating with substations during planned and emergency outages.
- Supporting restoration after system disturbances.
- Ensuring secure and uninterrupted power system operation.

Layout of the Control Centre

The SCADA Control Centre consists of several functional sections working together.

Operator Consoles

The operator consoles are equipped with multiple monitors displaying:

- Single Line Diagram (SLD)
- Voltage profile
- Frequency
- Power flow
- Transmission line status
- Transformer loading
- Alarm window
- Event logs

- Communication health

Each workstation enables operators to analyse real-time data and respond quickly to abnormal conditions.

SCADA Servers

The control centre contains dedicated SCADA servers responsible for:

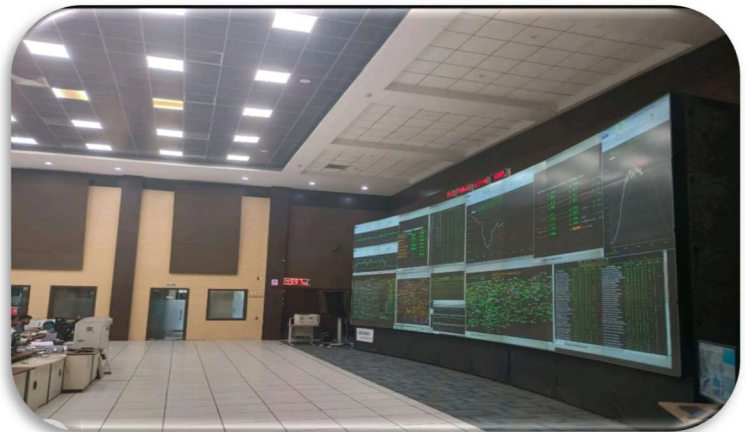
- Collecting data from substations
- Processing field information
- Storing historical records
- Managing alarms
- Communicating with EMS applications

To ensure uninterrupted operation, UPSLDC uses redundant server architecture, where a standby server automatically takes over if the primary server fails.

Human Machine Interface (HMI)

The Human Machine Interface (HMI) provides a graphical representation of the power system. Through the HMI, operators can monitor:

- Bus voltages
- Active power (MW)
- Reactive power (MVAR)
- Circuit breaker status
- Transformer loading
- Transmission line loading
- Frequency
- Equipment alarms



The HMI also allows authorized operators to issue remote control commands whenever required.

Information Displayed in the Control Centre

The SCADA Control Centre continuously displays the following operational parameters:

Electrical Parameters

- Voltage (kV)
- Current (A)
- Active Power (MW)
- Reactive Power (MVAR)
- Frequency (Hz)
- Power Factor
- Equipment Status

- Circuit Breaker Status
- Isolator Status
- Transformer Status
- Bus Coupler Status
- Line Availability
- System Status
- Total Generation
- State Demand
- Renewable Generation
- Inter-State Power Exchange
- Communication Health
- Network Alarms

Alarm and Event Management

One of the most important functions of the Control Centre is Alarm Management.

Whenever an abnormal condition occurs, the SCADA system immediately generates an alarm.

Examples include:

- Transmission line trip
- Transformer trip
- Breaker failure
- Low voltage
- High voltage
- Frequency deviation
- Communication failure
- RTU failure
- Battery charger failure
- Each alarm includes:
 - Time of occurrence
 - Equipment name
 - Location
 - Nature of fault
 - Priority level

This information enables operators to take timely corrective action.

Coordination During Grid Disturbances

In the event of a disturbance, the Control Centre coordinates with:

- Transmission Substations
- Power Generating Stations
- Distribution Companies (DISCOMs)
- Northern Regional Load Dispatch Centre (NRLDC)

Operators analyse the disturbance, identify the affected equipment, and guide field engineers to restore the system safely.

Practical Observations During Internship

During the internship, the following observations were made inside the SCADA Control Centre:

- A large digital video wall continuously displayed the real-time Single Line Diagram (SLD) of the Uttar Pradesh transmission network.
- Separate dashboards showed generation from Thermal, Hydro, Solar, and Independent Power Producers (IPPs).
- System frequency, state demand, and transmission line loading were monitored continuously.
- Engineers explained how alarms generated by SCADA help operators identify faults immediately.
- Communication health between substations and UPSLDC was monitored in real time to ensure uninterrupted data exchange.
- The Control Centre operated round the clock with engineers working in shifts to maintain continuous supervision of the grid.

Importance of the SCADA Control Centre

The SCADA Control Centre acts as the brain of the state transmission network. Without centralized monitoring, it would be extremely difficult to maintain the balance between generation and demand, detect faults quickly, coordinate restoration activities, and ensure the secure operation of the power system.

The combination of SCADA, EMS, advanced communication systems, and experienced engineers enables UPSLDC to provide reliable and efficient grid operation throughout Uttar Pradesh

The SCADA Control Centre is the operational hub of UPSLDC, where real-time monitoring, alarm management, communication supervision, and grid operation are carried out. The practical exposure during the internship provided valuable insight into how modern automation systems help maintain the stability and reliability of one of India's largest state power transmission networks.

Communication Infrastructure at UPSLDC

Introduction

A reliable communication system is the backbone of any modern Supervisory Control and Data Acquisition (SCADA) network. Without a secure and high-speed communication infrastructure, real-time monitoring and control of substations would not be possible.

At the Uttar Pradesh State Load Dispatch Centre (UPSLDC), an advanced communication network interconnects generating stations, transmission substations, and the SCADA Control Centre. This communication system enables continuous transmission of operational data, alarms, measurements, and control commands with high reliability and minimum delay.

During the industrial internship, the Communication Room was one of the most technically significant sections visited. It contained communication racks, optical fibre equipment, routers, switches, firewalls, GPS time synchronization devices, battery backup systems, and network monitoring software.

Importance of Communication Infrastructure

The communication infrastructure serves as the medium through which information travels between substations and the SCADA Control Centre.

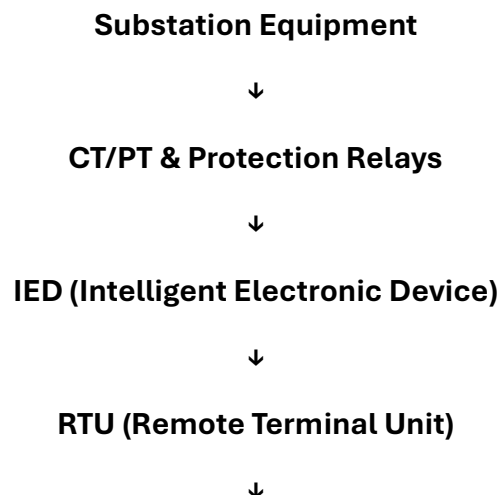
Its major objectives are:

- Reliable transfer of operational data.
- Continuous communication with substations.
- Fast transmission of alarms and events.
- Secure exchange of SCADA and EMS information.
- Remote monitoring and control.

High availability through redundant communication paths.

Communication Architecture

The communication architecture observed during the internship can be represented as follows:



Optical Fibre Communication (OPGW)



SDH / MPLS Communication Network



Routers & Network Switches



Fortinet Firewall



SCADA Servers



Operator Workstation (HMI)



SCADA Control Centre

This architecture ensures secure and uninterrupted communication between remote substations and UPSLDC.

Communication Room

The Communication Room houses all critical communication equipment responsible for transferring operational data throughout the transmission network.

During the internship, the following equipment was observed:

- Optical Communication Racks
- SDH Equipment
- Cisco Routers
- Managed Ethernet Switches
- Fortinet Firewalls
- GPS Time Synchronization Server
- Fibre Patch Panels
- SFP Modules
- Battery Charger System
- Network Monitoring Workstations

The communication room is designed with redundant architecture to ensure uninterrupted operation even in the event of equipment failure.

Optical Fibre Communication

UPSLDC primarily uses Optical Fibre Communication for transmitting operational data between substations and the Control Centre.

Advantages include:

- High bandwidth
- Low transmission loss
- High-speed communication
- Immunity to electromagnetic interference
- Reliable long-distance communication
- Enhanced cybersecurity

The optical communication network forms the primary communication backbone for SCADA operations.

OPGW (Optical Ground Wire)

The Optical Ground Wire (OPGW) combines the functions of a conventional earth wire and an optical fibre communication cable.

Major advantages:

- Lightning protection
- High-speed communication
- Low maintenance
- Long operational life
- Reliable communication between substations
- OPGW is extensively used across the Uttar Pradesh transmission network.

Network Switches

Managed Ethernet switches provide connectivity between communication devices.

Functions include:

- Data switching
- VLAN management
- Network monitoring
- Port management
- Communication redundancy
- Switches ensure efficient data transmission between SCADA devices.

Network Monitoring System

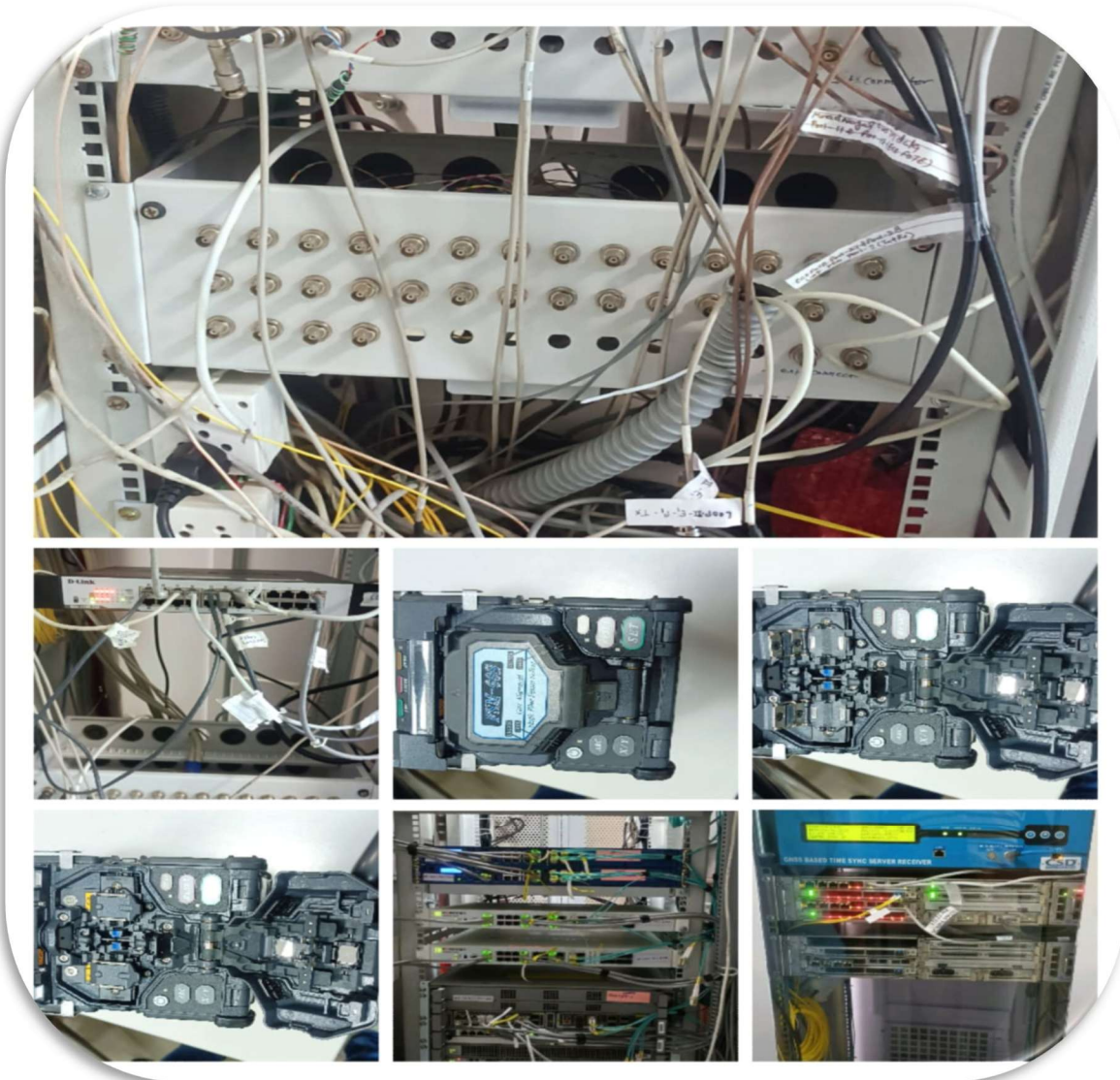
A dedicated Network Management System (NMS) continuously monitors the health of the communication network.

The system displays:

- Communication status

- Network topology
- Link availability
- Device status
- Alarm notifications
- Fault location

During the internship, the NMS screen displayed communication links connecting multiple substations and communication nodes across Uttar Pradesh.



Optical Ground Wire (OPGW)

Introduction

The Optical Ground Wire (OPGW) is one of the most important communication media used in modern electrical transmission systems. It combines the functions of a conventional earth wire and an optical fibre communication cable into a single conductor. OPGW is installed at the top of Extra High Voltage (EHV) transmission towers, where it provides lightning protection to the transmission line while simultaneously serving as the backbone for high-speed optical communication.

At Uttar Pradesh State Load Dispatch Centre (UPSLDC), OPGW forms the primary communication network connecting remote substations with the SCADA Control Centre. It carries real-time operational data, protection signals, voice communication, and Energy Management System (EMS) information, ensuring secure and reliable grid operation.

Need for OPGW

Modern power systems require continuous exchange of data between substations and the control centre. Traditional communication methods such as Power Line Carrier Communication (PLCC) have limitations in terms of bandwidth, speed, and reliability.

OPGW addresses these limitations by providing:

- High-speed communication
- Long-distance data transmission
- Reliable operation
- Lightning protection
- Low maintenance
- High availability

For these reasons, OPGW has become the preferred communication medium for utilities like UPSLDC.

Working Principle of OPGW

The OPGW performs two functions simultaneously:

1. Lightning Protection

Installed at the top of transmission towers, it intercepts lightning strikes and safely conducts the lightning current to earth through the transmission tower.

2. Optical Communication

The optical fibres inside the conductor carry:

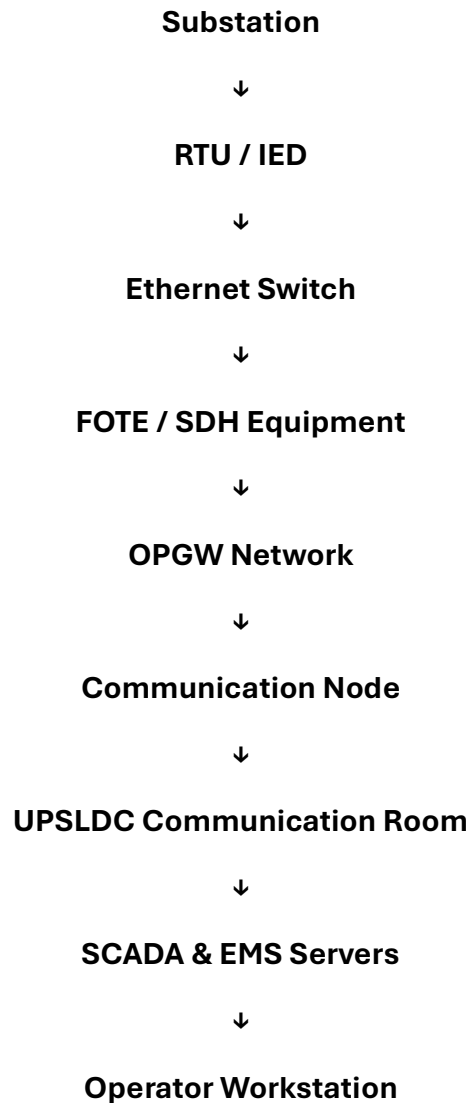
- SCADA data
- EMS data
- Protection signals

- Voice communication
- Network management traffic
- Time synchronization information

This dual functionality makes OPGW highly economical and efficient.

Communication Architecture Using OPGW

The communication architecture observed during the internship can be represented as:



The architecture enables secure and uninterrupted communication across the transmission network.

Applications of OPGW

At UPSLDC, the OPGW network supports multiple applications, including:

- SCADA communication
- Energy Management System (EMS)
- Teleprotection signalling
- Voice communication

OPGW and SCADA Integration

The OPGW network forms the communication backbone of the SCADA system. Operational data collected by RTUs at substations is transmitted through the OPGW network to the UPSLDC Control Centre.

The information transmitted includes:

- Voltage
- Current
- Frequency
- Active Power (MW)
- Reactive Power (MVAR)
- Circuit Breaker Status
- Transformer Status
- Alarm Signals
- Event Logs

This high-speed communication enables operators to monitor and control the transmission network in real time.

Practical Observations During Internship

Although the OPGW cable is installed on transmission towers and was not directly handled during the internship, its importance was explained by the engineers while studying the Communication Room and Fibre Optic Communication System.

It was observed that:

- Optical communication equipment in the communication racks interfaces with the OPGW network.
- SCADA and EMS data are transmitted through optical fibre links.
- Fibre patch panels and SFP modules provide connectivity between communication equipment and the optical network.
- Network health is continuously monitored through the Network Management System (NMS).
- Redundant communication paths are used to ensure uninterrupted operation in case of fibre faults.
- These observations highlighted the critical role of OPGW in maintaining reliable communication between remote substations and the State Load Dispatch Centre.



Protection System and Grid Security

Introduction

The protection system is one of the most critical components of an electrical power system. Its primary objective is to detect abnormal operating conditions, isolate faulty equipment, and ensure the continuous and reliable operation of the remaining healthy power network. A well-designed protection system minimizes equipment damage, improves system reliability, and reduces outage duration.

At the Uttar Pradesh State Load Dispatch Centre (UPSLDC), protection systems installed at generating stations and substations work in coordination with the SCADA, Energy Management System (EMS), Remote Terminal Units (RTUs), and Intelligent Electronic Devices (IEDs). Whenever a fault occurs, protection relays operate within milliseconds, trip the associated circuit breakers, and transmit fault information to the SCADA Control Centre for immediate action.

During the internship, engineers explained the importance of protection coordination and demonstrated how alarms and equipment status are monitored in real time through the SCADA system.

Objectives of the Protection System

The main objectives of a protection system are:

- Detect electrical faults rapidly.
- Isolate only the faulty section of the network.
- Protect expensive electrical equipment.
- Prevent cascading failures.
- Maintain grid stability.
- Ensure personnel safety.
- Minimize outage duration.
- Improve system reliability.

Types of Faults in Power Systems

Power system faults may occur due to insulation failure, lightning, equipment malfunction, tree contact, or human error.

The major types of faults are:

- Symmetrical Fault
- Three-Phase Fault (L-L-L)
- Although rare, this fault produces the highest fault current and is considered the most severe.
- Unsymmetrical Faults

These are more common and include:

- Single Line-to-Ground (L-G)
- Line-to-Line (L-L)
- Double Line-to-Ground (L-L-G)

Protection systems are designed to detect these faults quickly and accurately.

Components of a Protection System

A modern protection system consists of the following components:

- Current Transformer (CT)

Measures line current and provides a reduced current to protection relays.

- Potential Transformer (PT)

Measures system voltage and supplies a reduced voltage signal to protection devices.

- Numerical Protection Relay (IED)

Processes electrical measurements, detects faults, records events, and issues trip commands.

- Circuit Breaker (CB)

Interrupts fault current and isolates the faulty equipment.

- Remote Terminal Unit (RTU)

Communicates fault information to the SCADA Control Centre.

- SCADA & EMS

Display alarms, equipment status, and fault information for operator action.

Working of the Protection System

The protection process follows these steps:

- A fault occurs on the transmission line or equipment.
- CTs and PTs detect abnormal current and voltage.
- The numerical protection relay analyses the signals.
- If the measured values exceed the preset limits, the relay issues a trip command.
- The circuit breaker opens and isolates the faulty section.
- The RTU transmits fault information through the communication network.
- The SCADA system immediately displays alarms and breaker status.
- Operators at UPSLDC analyse the event and coordinate restoration.

Protection Operation Flow

Fault Occurs



CT / PT



Numerical Relay (IED)



Trip Command



Circuit Breaker Opens



RTU



SCADA



UPSLDC Control Room



Operator Action

Types of Protection Used in Transmission Systems

The following protection schemes are commonly used in transmission substations:

Overcurrent Protection

Protects equipment against excessive current caused by faults.

Earth Fault Protection

Detects leakage current flowing to earth due to insulation failure.

Differential Protection

Compares current entering and leaving protected equipment such as transformers. Any significant difference indicates an internal fault.

Distance Protection

Used mainly for transmission lines by calculating the apparent impedance between the relay and the fault location.

Overvoltage and Undervoltage Protection

Protects equipment against abnormal voltage conditions.

Frequency Protection

Operates when system frequency moves outside permissible limits.

Busbar Protection

Provides protection for busbars against internal faults.

Transformer Protection

Protects power transformers using differential, Buchholz, overcurrent, and temperature protection schemes.

Numerical Protection Relays

Modern substations use numerical relays instead of electromechanical relays.

Advantages include:

- High accuracy
- Faster operation
- Integration with SCADA and IEC 61850 networks

Role of SCADA in Protection

Although SCADA does not perform protection functions directly, it plays a vital role in monitoring protection operations.

SCADA displays:

- Breaker trip indication
- Relay alarms
- Fault location information
- Transformer status
- Transmission line status
- Event logs
- Time-stamped fault records

This information enables operators to take immediate operational decisions.

Grid Security

Grid security refers to maintaining stable, reliable, and uninterrupted operation of the electrical network under both normal and abnormal conditions.

At UPSLDC, grid security is ensured through:

- Continuous SCADA monitoring
- Energy Management System (EMS)
- Reliable communication network
- Protection coordination

Power Supply System at UPSLDC

Introduction

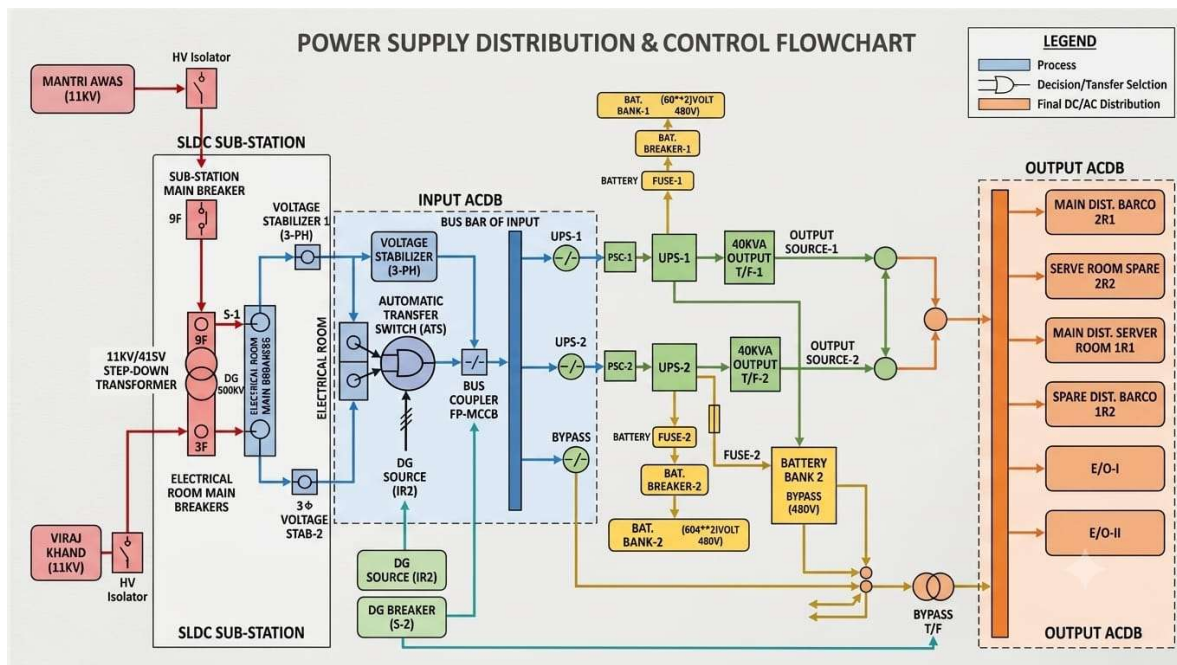
A reliable power supply system is one of the most critical infrastructures of the Uttar Pradesh State Load Dispatch Centre (UPSLDC). The uninterrupted operation of SCADA, Energy Management System (EMS), Communication Network, Protection System, Servers, RTUs, IEDs, Routers and other control equipment depends upon a highly reliable electrical power supply. Even a short interruption in supply can affect real-time monitoring, protection and grid control operations.

To ensure continuous operation, UPSLDC employs multiple levels of redundancy including dual incoming power sources, transformers, UPS systems, battery banks, diesel generator backup, bypass arrangements and dedicated AC/DC distribution systems.

During the industrial internship, the Electrical Room, UPS System, Battery Charger Room and Power Distribution System were studied to understand how uninterrupted power is supplied to critical equipment.

Overall Power Supply Architecture

The overall power supply arrangement observed at UPSLDC is illustrated below.



Incoming Power Supply

UPSLDC receives power through two independent 11 kV feeders to ensure uninterrupted operation.

During the internship, two incoming power sources were observed:

- Mantri Awas 11 kV Feeder
- Virajkhand 11 kV Feeder

The availability of two independent feeders increases system reliability. If one feeder becomes unavailable due to maintenance or fault, the second feeder continues supplying power.

Substation Transformer

The incoming 11 kV supply is stepped down to 415 V (0.4 kV) using a distribution transformer installed within the UPSLDC premises.

Functions:

- Voltage reduction
- Electrical isolation
- Reliable low-voltage supply
- Distribution to the Electrical Room

Electrical Room

The Electrical Room acts as the central power distribution area.

It contains:

- LT Panels
- Switchgear
- Circuit Breakers
- Bus Coupler
- Power Control Panels
- Protection Devices

Functions:

- Receives supply from transformer
- Controls incoming sources
- Distributes power
- Coordinates DG supply
- Supplies UPS system

Power Control System

The Power Control Panel selects the available source among:

- Source-A
- Source-B
- Diesel Generator

Its purpose is to maintain uninterrupted supply during source failure.

UPS System

The UPS (Uninterruptible Power Supply) provides continuous AC power to critical equipment.

During normal operation:



During power failure:



The UPS also:

- Charges batteries
- Filters voltage fluctuations
- Maintains constant output
- Eliminates interruptions

Two independent UPS units (UPS-1 and UPS-2) were installed to provide redundancy.

Battery Charger

The battery charger converts AC into regulated DC while simultaneously charging the battery banks.

Main functions:

- Float Charging
- Boost Charging

- Battery Monitoring
- Automatic Charging Control

Battery Bank

Battery banks store electrical energy.

Functions:

- Backup during power failure
- Supply DC power
- Operate protection systems
- Support UPS
- Supply control equipment

Battery banks ensure that no interruption occurs before the DG starts.

Rectifier Section

The rectifier converts AC into DC.

Major Components:

- Step-down Transformer
- Rectifier Bridge
- Filters
- Voltage Regulator

Output Transformer

After UPS, the output passes through Output Transformers.

Functions:

- Electrical Isolation
- Stable Output Voltage
- Improved Power Quality
- Protection of Sensitive Equipment

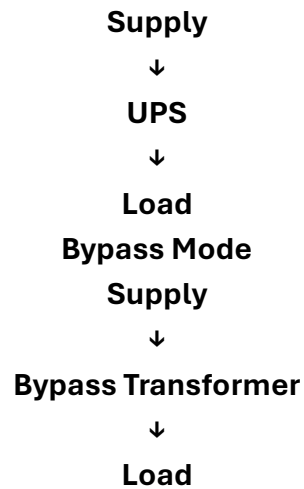
Two output transformers were observed:

- Output Transformer-1
- Output Transformer-2

Bypass System

A bypass system provides an alternate path during UPS maintenance or UPS failure.

Normal Mode



This allows maintenance without interrupting critical equipment.

AC Distribution Board (ACDB)

The ACDB distributes regulated power to:

- SCADA Servers
- EMS Servers
- Routers
- Firewalls
- Switches
- Communication Equipment
- Operator Consoles

Each feeder is protected by suitable protective devices.

Monitoring Parameters

The UPS system continuously monitors:

- Input Voltage
- Output Voltage

- Frequency
- Battery Voltage
- Charging Current
- Battery Health
- UPS Status
- Float Mode
- Boost Mode
- Alarm Status

Practical Observations

During the internship the following were observed:

- Dual incoming 11 kV supply
- Dedicated Electrical Room
- UPS-1 and UPS-2
- Battery Banks
- Battery Charger Panels
- Rectifier Modules
- Float and Boost Indicators
- Output Transformers
- Bypass Transformer
- AC Distribution Board
- Digital Monitoring Displays

The engineers explained that uninterrupted power is mandatory for SCADA, EMS, Communication Systems and Protection Equipment.

Advantages

- High Reliability
- Dual Power Sources
- UPS Redundancy
- Battery Backup

- DG Backup
- Continuous SCADA Operation
- Reliable Communication
- Improved Grid Security

Conclusion

The Industrial Internship at **Uttar Pradesh State Load Dispatch Centre (UPSLDC)**, Lucknow was a highly valuable and enriching learning experience that provided practical exposure to modern power system operation, automation technologies, industrial communication networks, and real-time grid management. It successfully bridged the gap between the theoretical concepts studied during the Bachelor of Technology programme and their practical implementation in the power sector.

Throughout the internship, I gained a comprehensive understanding of the Supervisory Control and Data Acquisition (SCADA) system, which serves as the backbone of real-time monitoring and control of the transmission network. I learned how operational data from substations is collected through Remote Terminal Units (RTUs) and Intelligent Electronic Devices (IEDs), transmitted through secure communication networks, processed by SCADA servers, and displayed on Human Machine Interface (HMI) workstations for continuous monitoring by system operators.

The internship also provided detailed knowledge of the Energy Management System (EMS) and its applications in generation scheduling, load monitoring, frequency control, voltage supervision, contingency analysis, and overall grid management. The coordination between SCADA and EMS demonstrated how advanced automation technologies contribute to the secure, reliable, and economical operation of the electrical power system.

A major highlight of the training was the practical exposure to the Communication Room, where industrial communication equipment such as Cisco Routers, Managed Ethernet Switches, Fortinet Firewalls, ABB Fibre Optic Communication Equipment, Meinberg GPS Time Servers, Fusion Splicer, Optical Time Domain Reflectometer (OTDR), SFP Modules, Battery Charger Panels, and Communication Racks were studied. This practical experience enhanced my understanding of communication infrastructure, fibre optic communication, Optical Ground Wire (OPGW), SDH networks, cybersecurity, and network redundancy.

The internship also strengthened my understanding of power system protection, generation scheduling, grid security, PMU and WAMS technologies, GPS time synchronization, and the importance of uninterrupted communication in maintaining the stability of the electrical grid.

In addition to technical knowledge, the internship improved my professional skills, including technical communication, analytical thinking, teamwork, documentation, observation, and disciplined work practices. Interaction with experienced engineers provided valuable insight into the responsibilities, decision-making processes, and operational challenges associated with managing a large state transmission network.

References

The following books, standards, technical manuals, and official publications were referred to while understanding the concepts presented in this internship report. These references are widely accepted in the fields of Power Systems, SCADA, Energy Management Systems (EMS), Communication Networks, Protection Systems, and Smart Grid Technologies.

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